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A nonlinear cable-like model of amplified ionic wave propagation along microtubules

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Abstract – The properties of ionic waves propagating along a microtubule in solution are analyzed in this paper. We derive the constitutive equations in the continuum limit, obtain appropriate parameter values from the tubulin structural data and solve numerically these equations finding their dependence on the range of parameter values. Possible biophysical implications of the results that may arise from the model are discussed.

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Microtubules (MTs) are cytoskeletal bio-polymers made up of GTP-dependent $\alpha\beta$ -tubulin dimer assemblies [1]. MTs are long hollow cylinders having outer diameters of 25 nm and inner diameters of 15 nm, with lengths that typically reach up to several micrometres. Recently, it has become apparent that neurons might utilise MT networks in cognitive processing via associated proteins (MAPs), including MAP-tau and MAP2, in neuronal processes such as learning and memory [2,3]. MTs are also linked to the regulation of a number of ion channels, thus contributing to the electrical activity of excitable cells [4]. Hence, MTs may play a role in the processing of electrical signals within the neuron. Structural reorganization of the cytoskeletal may reflect changes of neural circuits in response to learning and experience, and they must involve highly dynamic regulatory mechanisms. How this cytoskeletal control of neuronal plasticity takes form remains an open issue.

To elucidate the electrodynamic properties of MTs and gain insight into the regulatory role that MTs play in cellular activity, we assessed whether MTs are capable of processing electrical information via their effects on ionic conductivity in a buffer solution. More specifically, in this paper we present a model intended to explain the recent experimental observations showing an amplitude amplification of pulsed electric signals applied to individual MTs. In these experiments electrical data were

collected with a modified dual “patch-clamp” setup as previously reported [5]. Isolated taxol-stabilised MTs have been shown to be able to amplify an electric signal applied to them through a micropipette relative to the buffer solution. Amazingly, the signal that arrived at the other end of the MT was more than twice the signal recorded in a control experiment where the same two pipettes were immersed in a solution with no MT making contact with them. The calculated conductivity of MTs was found to be on the order of 10 nS, indicating a high level of ionic conductivity along an MT. This experimental observation indicates that MTs may, in addition to their well-documented structural role, play an important functional role in signalling at the subcellular level. This paper is intended to provide a physical model describing this potentially key functional role.

Molecular-mechanics calculations with tubulin structure [6,7], indicate a strong negative surface charge distribution on the order of 20–25 e[−] per monomer, depending on the type of tubulin used, distributed preferentially on the outer surface over the inner core at a charge ratio of 2:1. Microtubules, therefore, being highly charged polyelectrolytes, are expected to contain a fraction of their surrounding counterions in the form of a condensed cloud localised around its surface (see fig. 1).

Based on Manning’s theory of polyelectrolytes causing ionic condensation along the stretch of a polymer [8], if a sufficiently high linear electric-charge density is present on the polymer’s surface, we assess that the cylindrical volume of depleted ions outside the ionic

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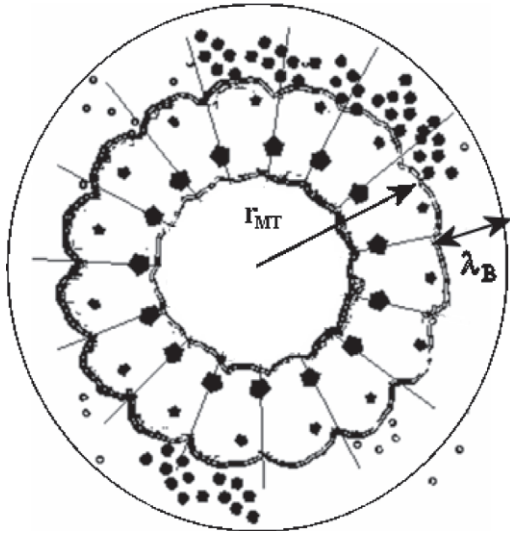


Fig. 1: Cross-section of the microtubule showing the condensed layer of width λ_B .

cloud surrounding the MT serves as an electrical shield. Thus, the “cable-like” behaviour of such a structure is supported by the polymer itself and the “condensed” counterions, which for practical purposes are “bound” to the polymer. The strength of this interaction is such that even under infinite dilution conditions, the counterions are still attracted to the polymer and do not diffuse out from its vicinity. Although this theory was originally postulated for such polyelectrolytes as DNA, the same applies to highly charged quasi-one-dimensional and cylindrical polymers such as actin and MTs.

Since our hypothesis is crucially dependent on the condensation of ions around the polymer’s cylindrical length (see fig. 1), we briefly evaluate the effective density of charges along a MT. The Bjerrum length, λ_B , describes the distance beyond which thermal fluctuations are stronger than the electrostatic attraction or repulsion between charges in solution whose dielectric constant is ϵ . It is defined by $\lambda_B = e^2 / (4\pi\epsilon_0 k_B T)$ for a given absolute temperature T . Here e is the elementary electric charge, ϵ_0 the permittivity of vacuum, and k_B is Boltzmann’s constant. For a temperature of 293 K, $\lambda_B = 7.13 \times 10^{-10}$ m. Counterion condensation occurs when the mean distance between charges, b , is such that $S = \lambda_B / b > 1$. We assume that each beta-tubulin monomer carries an excess of 24 negative charges in vacuum while alpha carries 22 charges. Typically 40% of the net charge is located on surfaces of tubulin that are not exposed to the solution. Consequently, we expect approximately 18 negative charges per dimer excluding the C-terminal tails which carry about 10 charges. Assuming an 8 nm length per dimer and 13 protofilaments forming a microtubule, we estimate that there is ~ 30 e/nm. Thus, we expect a linear charge spacing $b = 0.34 \times 10^{-10}$ m, so $S \approx 21$. As the effective charge, q_{eff} ,

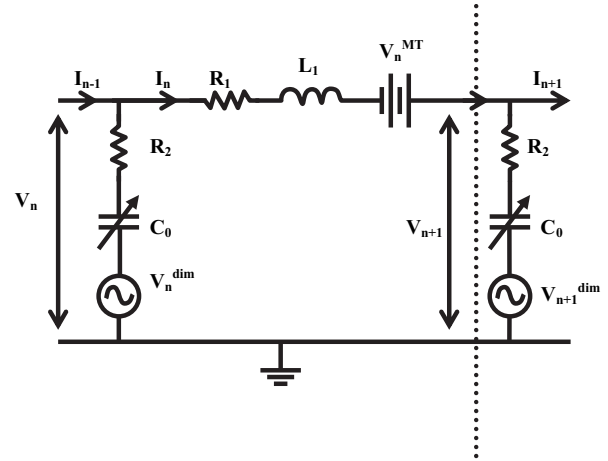


Fig. 2: A diagram describing the electronic analog of the microtubule model.

or renormalised rod charge is the bare value divided by S , we find $q_{\text{eff}} \approx 0.43$ e/monomer. This value is reasonably close to the one measured experimentally in assays where microtubules were subjected to external electric fields and their gliding mobility was measured leading to an estimate of 0.15 e/dimer. However, this represents a “far-field” value where additional screening is achieved by ions attracted to the MT but not necessarily condensed on it. Bearing in mind the sheath of counterions around the microtubule cylinder, we see that effectively MTs may act as biological “electrical wires”, which can be modelled as nonlinear inhomogeneous transmission lines capable of propagating nonlinear dispersive extended or solitary waves [9] (and references therein).

To investigate the ionic-conduction properties of an MT we developed a nonlinear electrical circuit, which mimics the behaviour of an MT in solution. As the cylinder’s core is separated from the rest of the ions in the bulk solution by the counterion condensation cloud, we expect this cloud to act as a dielectric medium between the two. It provides resistive and capacitive components for the behaviour of the dimers that make up the MT. Ion flow is expected at a radial distance from the centre of the cylinder, which is approximately equal to the Bjerrum length. An inductive component to the electrical properties of ionic waves is due to the helical MT structure that, like a solenoid, should induce a helical ion flow. Below we discuss the origin of each electrical component in turn following a model developed earlier for an actin filament [9]. The full derivation of the model equations is beyond the scope of this letter and will be given elsewhere.

In order to derive the model equation, we begin by introducing the discrete potential in one section of the MT where Kirchhoff’s laws for the currents and voltages are applied, fig. 2. The idea is to develop a spatio-temporal equation for the potential over the capacitance as a function of the parameters of the model. The voltage across the “input” to the section, v_n , may be written in

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the following form:

$$v_n = R_2(I_{n-1} - I_n) + V_n^C + v_n^{\text{dim}}, \quad (1)$$

where V_n^C is the potential over the capacitance, therein denoted by $V_n = V_n^C$. The relation between the current entering segment AB and the current across the capacitance is given by

$$I_{n-1} - I_n = \frac{dQ_n}{dt}. \quad (2)$$

The next relation arises from the potential difference in segment AC:

$$v_n - v_{n+1} = L \frac{dI_n}{dt} + R_1 I_n + V_n^{\text{MT}}. \quad (3)$$

Taking into account the potential over the effective solenoid and the permanent potential from the dimer's atomic structure, we expand the last equation in terms of the axial current, and differentiating it once we obtain

$$L \frac{d^2 Q_n}{dt^2} = (v_{n-1} - 2v_n + v_{n+1}) + R_1(I_n - I_{n-1}) + V_n^{\text{MT}} - V_{n-1}^{\text{MT}}. \quad (4)$$

Based on previous analysis of the electric-field distribution around the dimer and its permanent dipole moment (*e.g.* [7]) we assume that the local contribution of each dimer is of the form $v_n^{\text{dim}} = V_0 \sin(kna) + u_0$ and the linear longitudinal contribution due to the constant dipole takes the form $V_n^{\text{MT}} = \alpha n + u'_0$, where V_0 , u_0 , u'_0 , and α are constants, n is the discrete segment index of the MT, and a is the typical length. The wave number k describes the periodicity of the surface potential.

The capacitance in our model represents the charge distribution in the region from the surface of the MT to approximately one Bjerrum length away perpendicularly to it. Since the ions in this layer are assumed to be condensed, the distribution of charges will be nonlinear. For simplicity we assume a simple exponentially decaying nonlinearity, where taking an expansion up to the second order results in the following relation:

$$Q_n = C_0 (V_n - bV_n^2). \quad (5)$$

Following a similar mathematical exercise as in [9], and applying the continuum limit to the discrete equation we obtain an equation for the potential V for small spatial and temporal variations, as well as small nonlinearity b :

$$LC_0 \frac{d^2 V}{dt^2} = a^2 V'' - R_1 C_0 \frac{dV}{dt} + 2R_1 C_0 b V \frac{dV}{dt} + R_2 C_0 \frac{d}{dt} (a^2 V'' - V_0 (ka)^2 \sin(kx) + \alpha). \quad (6)$$

This last equation is a nonlinear 3rd-order differential equation with derivatives in time. In order to facilitate the analysis we focus on a special family of solutions which

represent a propagating wave with a constant velocity. To achieve this, we use the coordinate transformation, $\xi = x - vt$, so the potential becomes a function of a single variable, $V(x, t) \rightarrow V(\xi)$:

$$LC_0 v^2 V'' = a^2 V'' + v R_1 C_0 V' - 2R_1 C_0 b v V V' - R_2 C_0 v a^2 V^{(3)} - V_0 (ka)^2 \sin(k\xi) + \alpha, \quad (7)$$

where $V' = dV/d\xi$. This 3rd-order equation can be integrated once to give

$$R_2 C_0 v a^2 V'' + (LC_0 v^2 - a^2) V' - v R_1 C_0 V + R_1 C_0 b v V^2 = V_0 (ka)^2 \cos(k\xi)/k + \alpha \xi + \text{const} \quad (8)$$

which brings us to a final 2nd-order nonlinear equation. Since this equation does not have a closed analytic solution, we perform a numerical analysis of its properties.

The circuit component values are derived from their physical properties. The capacitive component in our model represents the nonlinear condensed layer perpendicular to the MT surface. Hence, the linear part of the capacitor is that of a cylindrical layer whose inner radius equals that of the microtubule. For a cylinder of length l the capacitance would be

$$C_0 = \frac{2\pi\epsilon_0 l}{\ln[(r_{\text{MT}} + \lambda_B)/r_{\text{MT}}]}. \quad (9)$$

Using the parameters $r_{\text{MT}} = 12.5$ nm, $\lambda_B = 1$ nm, and $l = 8$ nm, we obtain $C_0 \approx 6.6 \times 10^{-4}$ pF.

The resistive element is composed of two contributions: one parallel to, and one perpendicular to the main MT axis. The axial contribution, R_1 , is basically the same as for actin and we shall not repeat its derivation. The perpendicular contribution arises from the resistivity between the two concentric cylinders:

$$R_2 = \frac{\ln[(r_{\text{MT}} + \lambda_B)/r_{\text{MT}}]}{2\pi l \sigma}, \quad (10)$$

where σ is the ionic solution conductivity. For physiological solutions, conductivity is governed mainly by the presence of sodium and potassium ions, which gives $\sigma \approx 1.2 (\Omega\text{m})^{-1}$. Hence the perpendicular resistance becomes $R_2 \approx 1.2$ M Ω which is approximately one-fifth of the value obtained for actin.

The solenoidal element stems from the intrinsic helical structure observed in MTs. This characteristic is a result of the lateral translation between adjacent protofilaments. We use the following relation for the inductance: $L = \mu N^2 A/l$, where N is the effective number of turns per dimer, μ is the magnetic permeability, A is the cross-sectional area and l is the length of the element, taken here as the length of three monomers. The effective number of turns is roughly approximated here as the number of ions that can be arranged along the protofilament, taking into account the hydration shell — $r_h \approx 3.6$ Å. Substituting these relations we obtain

$$L = \frac{\mu (l/r_h)^2 \pi (r_{\text{MT}} + \lambda_B)^2}{l}. \quad (11)$$

Applying the parametric values we find $L \approx 30$ pH.

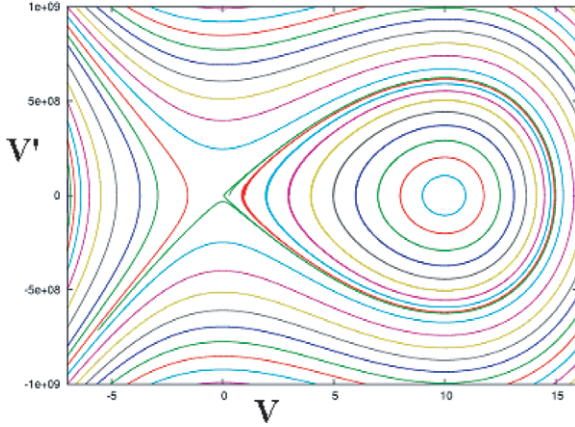


Fig. 3: Solutions in phase space. In the egg-shaped region trajectories circulate clockwise eventually moving far enough outwards and escape to $(-\infty, -\infty)$.

The 2nd-order equation derived above in a moving framework, eq. (8), has no general analytical solution; however, the equation has been analysed numerically. Having transformed our second-order nonlinear differential equation into a system of two coupled equations, we solved it numerically for various initial conditions. Plotting some of these in phase space produces fig. 3, where the first-order derivative, V' , is depicted *vs.* the potential V .

To further study the amplification character of the obtained solutions, we investigated the solutions with trajectories in the egg-shaped interior region of fig. 3. Here, trajectories passing through $(V, 0)$ at $\xi = 0$ are considered. If these were closed paths, this point would be the minimum V , instead these are only local minima and on each cycle the trajectory spirals outward to successively lower local minima until finally escaping the interior region. For the trajectory passing through $(9.9, 0)$ at $\xi = 0$ this takes over 1500 cycles and occurs at ξ of about $14000a$. Details of these process are shown in figs. 4a and b. This demonstrates the functional form of the solutions that describe the amplification process.

We should also discuss the issue of the thickness of the condensed layer that has been the subject of recent papers [10,11] in the context of the Onsager-Manning-Oosawa counterion condensation theory. While in our calculations, the Bjerrum length was used to determine the spatial extent of the counterions congregating around the filament, these papers suggest the use of the ‘‘Manning radius’’. We have applied the analysis presented in [11] and got a Manning radius ~ 6 times the Debye length, or about 6 nm. This is almost 9 times the Bjerrum length. Subsequently, the values of the circuit parameters have been recalculated and we obtained, a fivefold reduction in C_0 , a fivefold increase in R_2 , and a twofold increase in L . However, while the quantitative details of the problem have been altered the qualitative nature of the solutions’ behaviour has not, as is shown in the revised phase diagram in fig. 5.

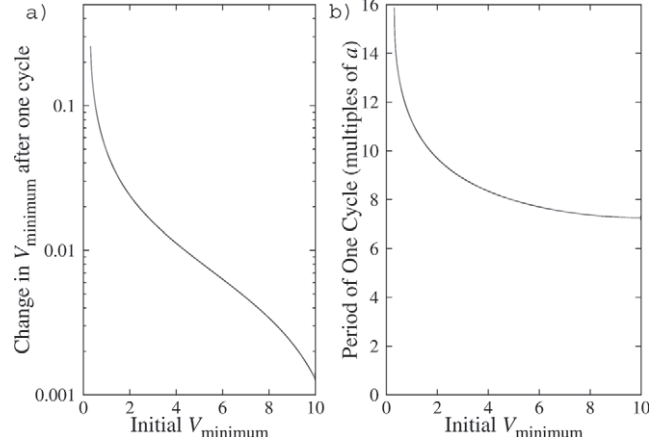


Fig. 4: Cycles in the interior show an amplification process. a) The decrease in the local minima (crossing $V' = 0$), each cycle for different starting positions. Note that at V near 10 the cycles are more stable. Similar graphs may be obtained using other ‘‘landmarks’’ in the cycle such as the maxima in V or the extrema in V' ; however, the change curves are significantly different in these cases. b) Times taken to complete one ‘‘orbit’’ and return to $V' = 0$ near the starting point as a function of the initial position.

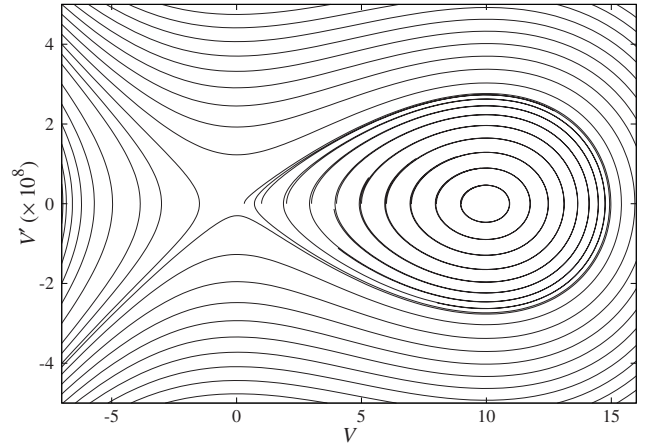


Fig. 5: Solutions in phase space for eq. (8) after changing the model parameters to account for the Manning length instead of the Bjerrum length. In the egg-shaped region trajectories circulate clockwise eventually moving far enough outwards and escape to $(1, 1)$. Note the similarity with fig. 3 that was obtained for the same equation but with the parameters calculated based on the Bjerrum-length concept.

Experimenting with omitting the sinusoidal term suggests that it is less significant. Omitting the first derivative term has a noticeable effect. The outward spiral in the egg-shaped region of fig. 3 was reduced. Qualitatively, the obtained solutions represent oscillations with an increasing amplitude. The separatrix region with a cusp at the origin, $(0, 0)$, represents an unstable solution. It is interesting to note that when the additional potential terms vanish, *i.e.* $V_0, \alpha \rightarrow 0$, we recover a similar equation as for the actin case [9] and hence we should expect a

solitary wave solution. However, in view of the above analysis, this solution is unstable, therefore hard to obtain numerically. So, we conclude that our MT equation should also support such solutions under similar conditions.

The model presented here is similar to the earlier model proposed for ionic propagation along actin filaments [9] but it differs by accounting for a modulated electrostatic potential along the filament's surface and a potential gradient between the MT ends. Also, specific model parameters have been selected for the MT case. It should be mentioned that in ref. [12] the problem of diffusion between two probes/electrodes in electrolytic solution has been discussed demonstrating the importance of a weakly nonlinear approximation consistent with our approach. However, the analysis in this latter paper is for long-range diffusion and not wave propagation. In addition, the derivation is for large potential differences, much higher than assumed for biological environments. We were mainly concerned with the propagation along the polymer in a direction approximately tangent to the long axis, and less so in the perpendicular direction. The wave propagation is confined by the radial direction to the condensed layer.

In summary, we have developed a biophysically based model of ionic conduction along MTs that gives qualitative support for the observations of ionic signal propagation along MTs that exhibited amplitude amplification. The existence of possibly unstable solitary wave solutions is predicted and should be the subject of future experimental investigations. The biophysical significance of this work is that it presents a nonlinear model of ionic conductivity along charged polar biopolymers such as MTs and shows how it may lead to signal amplification along particular directional pathways. This is a crucial issue in our understanding how subcellular signals can propagate amidst large amounts of thermal noise in an aqueous environment.

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